PS Analysis in Data Science: Problem Set 13

Problem 1. Let $f : (0, \infty)^n \to \mathbb{R}$, given by

$$f(x) = x_1 x_2 \cdots x_n,$$

and consider the constraint manifold

$$M = \{x \in \mathbb{R}^n : x_1 + \cdots + x_n = n\}.$$

(a) Show that M is an $(n-1)$-dimensional submanifold of $\mathbb{R}^n$.

(b) Show that every constrained extremum $x \in (0, \infty)^n$ satisfies

$$\nabla f(x) = \lambda(1, \ldots, 1).$$

(c) Prove that

$$f(x) \leq 1, \qquad \forall x \in M \cap (0, \infty)^n,$$

and determine when equality occurs.

Problem 2. Consider $f : \mathbb{R}^3 \to \mathbb{R}$,

$$f(x, y, z) = e^x + y^2 + z$$

subject to the constraints

$$g_1(x, y, z) = x + y + z - 1 = 0,$$
$$g_2(x, y, z) = x^2 + y^2 - z = 0.$$

(a) Show that the feasible set $M = \{(x, y, z)^\top \in \mathbb{R}^3 : g_1(x, y, z) = 0, \; g_2(x, y, z) = 0\}$ is a 1-dimensional submanifold.

(b) Derive the Lagrange multiplier system

$$\nabla f(x, y, z) = \lambda \nabla g_1(x, y, z) + \mu \nabla g_2(x, y, z), \qquad g_1(x, y, z) = 0, \qquad g_2(x, y, z) = 0.$$

(c) Apply Newton's method to find local extrema of f restricted to M.

Problem 3. Consider $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x) = x_1^2 + 2x_2^2$$

subject to the constraint

$$g(x) = x_1^2 + x_2^2 - 1 = 0.$$

Use the Lagrangian and its Hessian to identify the local minimum of the restriction of f. Visualize your result in Julia.

Problem 4. Consider $f : \mathbb{R}^2 \to \mathbb{R}$ given by

$$f(x, y) = x^2 + 2xy + 5y^2.$$

(a) Show that f has a unique global minimizer.

(b) Write the gradient descent iteration

$$u_{k+1} = u_k - \tau \nabla f(u_k), \qquad u_k = (x_k, y_k)^\top, \qquad \tau > 0,$$

in the form

$$u_{k+1} = A_\tau u_k,$$

where $A_\tau \in \mathbb{R}^{2\times 2}$.

(c) Determine all step sizes $\tau > 0$ for which $u_k \to 0$ for every initial vector $u_0 \in \mathbb{R}^2$.